

Course Overview

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Overview

- Big Data Techniques & Technologies: 4 credits
  - 20h lectures
  - 16h discussions & exercises
  - 24h projects

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Motivation

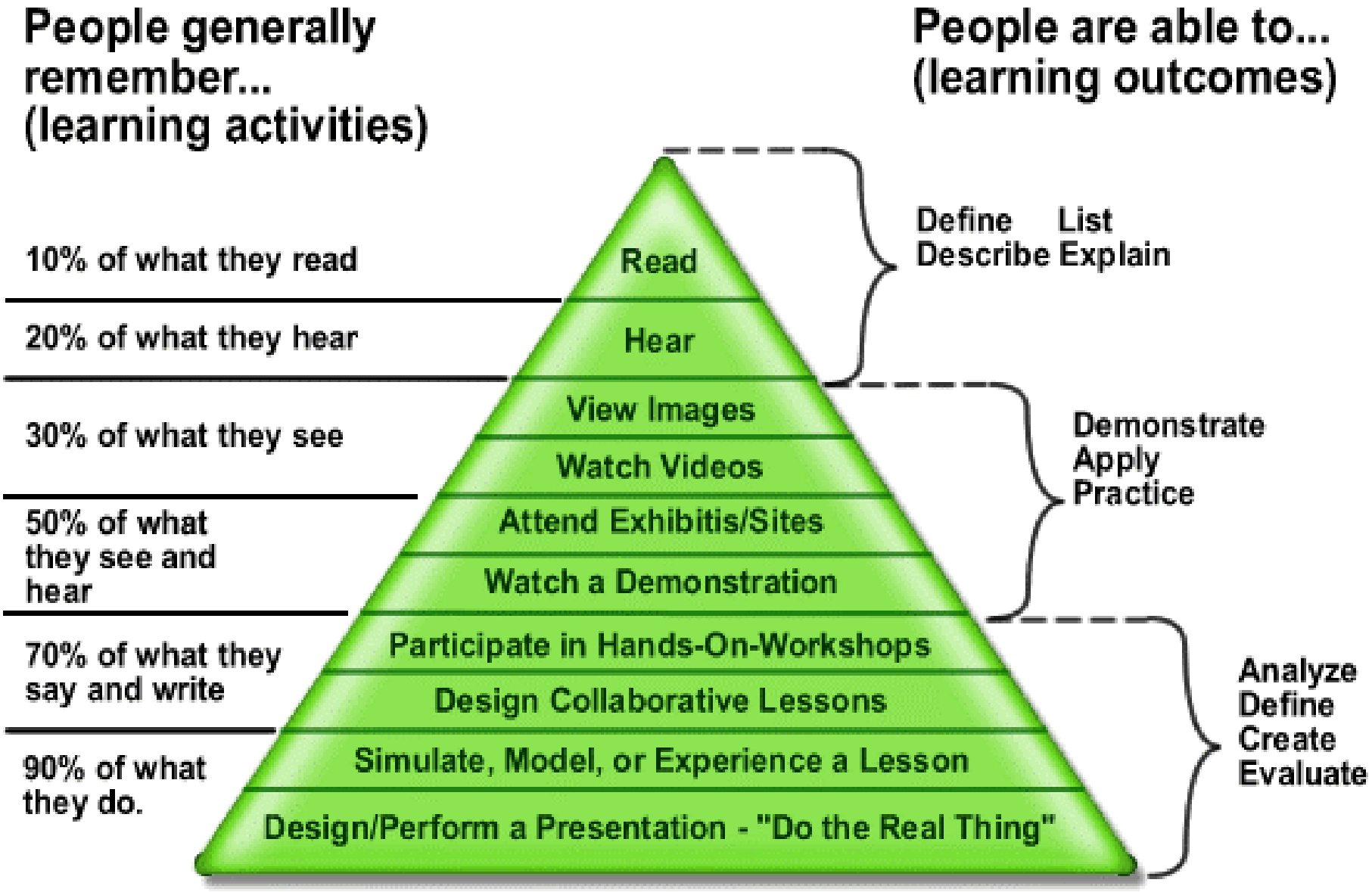
- Big data is everywhere
- A fundamental goal across numerous modern businesses and sciences is to be able to utilize as many machines as possible, to consume as much information as possible and as fast as possible.
- The big challenge is "***how to turn data into useful knowledge"***, moving target as both the underlying hardware and our ability to collect data evolve.

Objectives & Outcomes

- Aim to design, evaluate, and implement scalable distributed data systems that transform massive data into actionable intelligence.
- Students will be able to:
  1. **Understand and apply fundamental concepts**: Comprehend the foundational principles of big data management and analytics (structure of the big data ecosystem, data life cycle, …)
  2. **Leverage Big Data Programming Paradigms**: Employ key big data programming paradigms—such as streaming, parallel computing, and MapReduce—to develop scalable and efficient solutions.
  3. **Evaluate Modern Systems and Trends**: Analyze state-of-the-art research and industry trends in big data systems, and understand the trade-offs involved in designing and implementing these systems.
  4. **Make Informed Design Decisions**: Apply critical thinking to make strategic design decisions in data-driven scenarios, balancing performance, scalability, and system complexity.

Course Objectives & Outcomes...

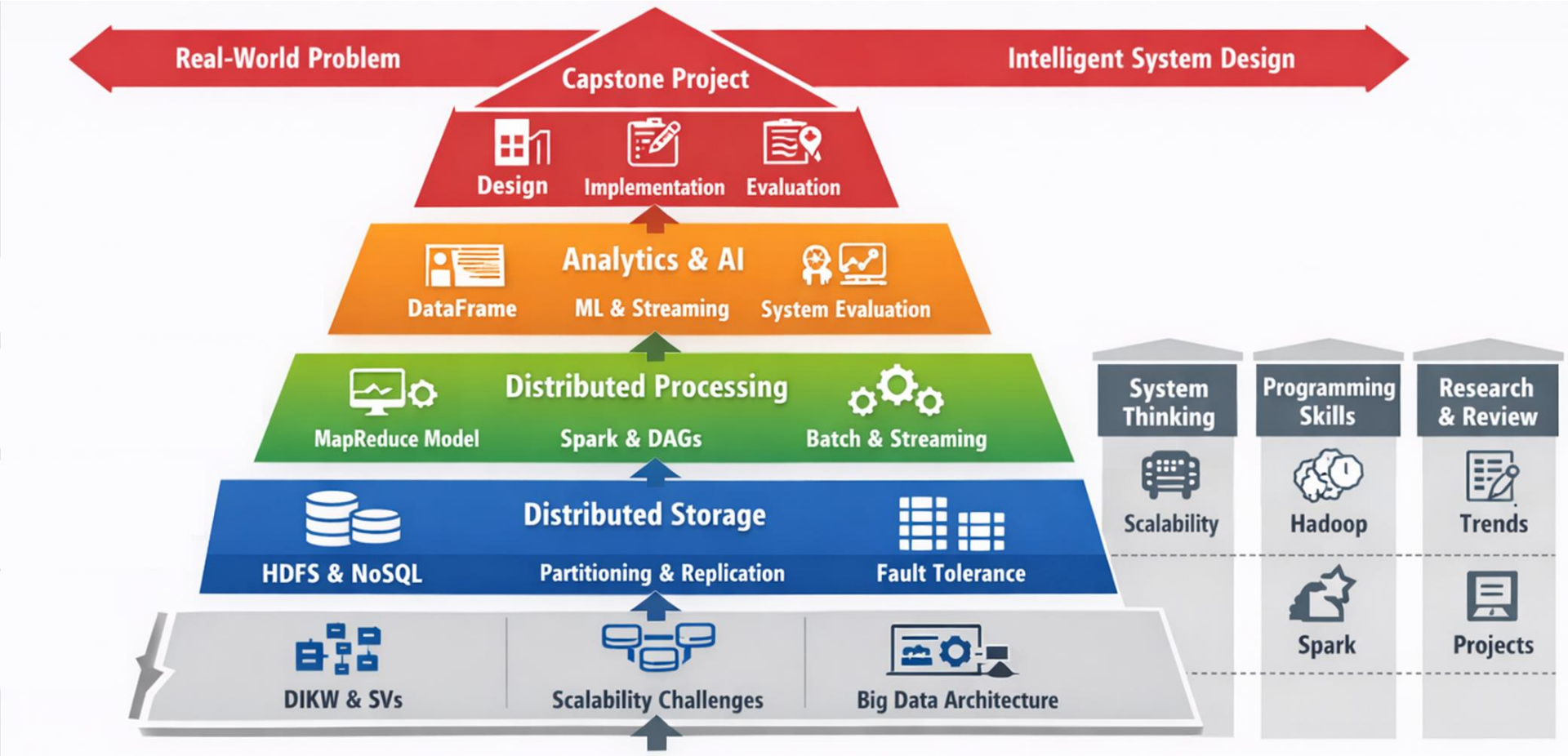
5. **Integrate Diverse Technological Principles**: Connect and utilize the fundamental principles across diverse areas—such as SQL, NoSQL, neural networks, graphs, statistics, data science, LLMs, and computer vision—in the context of big data.
6. **Develop Research Competencies**: Cultivate essential research skills including reading, writing, and critically evaluating research papers in the field of big data.
7. **Enhance Big Data Programming Proficiency**: Deepen programming expertise in Python with a focus on debugging, performance profiling, and developing robust analytical solutions.
8. **Solve Real-World Big Data Challenges**: Gain practical experience in identifying complex big data challenges and formulating effective, analytical solutions.


People generally remember... (learning activities)
People are able to... (learning outcomes)
10% of what they read
20% of what they hear
30% of what they see
50% of what they see and hear
70% of what they say and write
90% of what they do.
Read
Hear
View Images
Watch Videos
Attend Exhibitis/Sites
Watch a Demonstration
Participate in Hands-On-Workshops
Design Collaborative Lessons
Simulate, Model, or Experience a Lesson
Design/Perform a Presentation - "Do the Real Thing"
Define List
Describe Explain
Demonstrate
Apply
Practice
Analyze
Define
Create
Evaluate

Tentative Schedules

- S1: Course Overview
- S2-3: Introduction: Foundations of Big Data & Big Data Ecosystem
- S4-5: Big Data Storage Solution: SQL/NoSQL/NewSQL, Object Storage, and HDFS
  - Lab 1: Hadoop Cluster Setup & Configuration
- S6-7: Big Data Processing Paradigms: MapReduce
  - Lab 2: Hadoop-based Wordcount
- S8-9: Spark- Unified Engine for Large-Scale Data Analytics
  - Lab 3: Spark-based Wordcount
- S10-11: Data Analysis with Dataframe
  - Lab 4: Big Data Analytics
- S12-13: Big Data Framework/Platform/Solution Reviews
- S14-20: Capstone Project: big data in finance, health, education, environment, smart city, government, cybersecurity, ITS, GenAI, data platforms, social network analysis, recommendation, …

Conceptual Learning Map



Your feedbacks are important

- Big Data is a new topic, and thus the course is tentative
- The technologies keep evolving, and the course materials need to be updated correspondingly
- Please advise where I can improve after each lecture, at the discussion and QA.

Grading

Labs	30%	Programming assignments using big data frameworks, research experiments.
Participant	10%	Proactive interaction in the course.
Paper/System Reviews	20%	Reviewing research papers or assessing big data systems / frameworks.
Project	40%	Report (~15p) + Presentation (60”)

Textbooks

1. Recent research papers on big data topics.
2. Hadoop: The Definitive Guide, by Tom White. O'Reilly Media, 4th Edition, 2015.
3. Beginning Apache Spark 3: With DataFrame, Spark SQL, Structured Streaming, and Spark Machine Learning Library, Apress (2021) by Hien Luu.
4. Learning Spark: Lightning-Fast Big Data Analysis, 2nd Edition, O'Reilly, 2020, by H. Karau, A. Konwinski, P. Wendell, and M. Zaharia
5. Data Science and Big Data Analytics (John Wiley & Sons, Indianapolis IN, 2015) by EMC Education Services
6. Learning Elastic Stack 7.0: distributed search, analytics, and visualization using Elasticsearch, Logstash, Beats, and Kibana (2019) by Sharath Kumar M. N. and Pranav Shukla.

Course Project Overview

- The course project aims to:
  - Apply theoretical knowledge to a real-world technical problem
  - Develop teamwork, research, and implementation skills
  - Deliver a working prototype and technical report
- Team formation
  - Maximum 5 students per group
  - Topic will be finalized after Week 3
  - Early discussion and idea exploration are encouraged
- Project scope
  - Research-based and implementation-oriented
  - Requires technical design, experimentation, and evaluation
  - Final deliverables: prototype + demo/presentation + report

Technical Requirements & Preparation

- Hardware / Cloud Requirements: Each group must have:
  - At least one high-performance machine: CPU > 10 cores; RAM > 16GB; Storage > 512GB, or
  - A valid Google Cloud / Microsoft Azure / AWS account
- Required Technical Skills: Students should be comfortable with:
  - Use of Java and the Python data science ecosystem
  - Linux environment (Ubuntu 24.04 LTS) with basic command-line operations (to employ in virtual clusters for simplicity)
  - Visual Studio Code + SSH Extension, using Jupyter as an integrated environment
  - Packages: Spark for parallel programming, SciPy, Pandas, PySpark, …
- Preparation Recommendation
  - Install and configure development environment early
  - Test SSH remote development workflow
  - Ensure cloud credits/access before Week 3

Course Philosophy

- Rely on the latest research papers instead of a standard textbook,
- Lectures based on interaction and discussion instead of just “lecturing”,
- Attendance in lectures, labs and office hours is optional. The best way to learn, though, is through discussion and interaction with the instructors,
- NO LAPTOP/PHONE POLICY during the lecture.

How to Succeed

- Do the weekly readings (see syllabus)
- Take notes associated with the lectures.
- Practice exercises in the back of each reading.
- Attend class and actively participate.
- Begin assignments early and seek help if trouble (e.g. office hours).